

ethanol blending. The Centre told Parliament in April that between ESY 2014-15 and February 2026, it enabled faster payment of ₹1.5 lakh crore to farmers, saved over ₹1.7 lakh crore in forex, reduced carbon dioxide emission by 86.9 million tonnes and substituted over 28.9 million tonnes of crude oil.

Water and energy experts, however, advise caution against short-term gains.

K.J. Joy of Pune-based non-profit Society for Promoting Participative Ecosystem Management, notes: "Large-scale diversion of crop land for ethanol feedstock will definitely impact food and water security" and, hence, India needs an integrated approach to energy keeping food and water at its core.

Gopal Sarangi of TERI School of Advance Studies says: "The trade-off between food and energy security needs a fine balance as it involves competing demands on food, fodder, water and land. More so since ethanol blending is highly profitable for farmers and industry." India must find this balance at national or state levels, he adds, and that requires a database to determine the threshold beyond which the balance will be disturbed.

Sarangi accords higher priority to food security by arguing that India has multiple green energy choices.

THE ALTERNATIVES

It's not that the choice in India is only between imported crude oil and gas and ethanol. Apart from alternatives like renewable energy, nuclear power, and green hydrogen, there are options even in ethanol.

Debasis Das, Professor at the Indian Institute of Technology, Guwahati, points to second- and third-generation (2G and 3G)



DEBASISH DAS
PROFESSOR, IIT GUWAHATI

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CK JAIN
PRESIDENT, GRAIN
ETHANOL MANUFACTURERS
ASSOCIATION

"Before grains were allowed, the industry faced shortages as sugarcane supply was not sufficient. Now... we have sufficient feedstock and installed capacity to meet over 30% blending target"

ethanol plants that use non-food feedstock, eliminating risks to food security.

Three 2G ethanol plants are operational currently: Indian Oil's Panipat and Bharat Petroleum's Bargarh plants use paddy straw and Numaligarh Refinery uses bamboo. Hindustan Petroleum is building another at Bathinda to use paddy straw. Indian Oil also runs a 3G plant at Panipat that uses refinery off-gas.

Das tells *Business Today*: "These are exciting developments that take the load off grains. We need more 2G and 3G plants. We must also develop technology for other non-food feedstocks like rice husk, wheat straw, bagasse, Napier grass and wood waste."

To be sure, some other avenues have been tested in the country. Maharashtra made bamboo blending in thermal plants mandatory in 2025; it has approved a bamboo-to-methanol plant in Chandrapur. Two private firms produce green electricity and hydrogen through coal gasification.

At least 50 compressed biogas plants use agriculture residue, cattle dung, municipal solid waste, etc., for blending with CNG and PNG. Attempts are on to substitute imported palm stearin oil with used cooking oil for bio-diesel blending.

India need not rush with higher ethanol blending and create new vulnerabilities. Its role model, Brazil, started on this journey in 1975, and took 50 years to make E30 mandatory in 2025, despite rolling out E100-compliant cars in 1979. Though launched in 2003, India's ethanol blending began about a decade ago.

The country must tread cautiously and explore all options to ensure that addressing one vulnerability does not create others. **BT**